

- 3 A curve  $C$  has equation  $y = \frac{ax^2 + x - 1}{x - 1}$ , where  $a$  is a positive constant.

(a) Find the equations of the asymptotes of  $C$ .

[3]

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(b) Show that there is no point on  $C$  for which  $1 < y < 1 + 4a$ .

[4]

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- (c) Sketch  $C$ . You do not need to find the coordinates of the intersections with the axes. [3]